



US012402940B2

(12) **United States Patent**
Cosmescu

(10) **Patent No.:** **US 12,402,940 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **ULTRAPOLAR ELECTROSURGERY BLADE
AND ULTRAPOLAR ELECTROSURGERY
PENCIL**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **I.C. Medical, Inc.**, Phoenix, AZ (US)
(72) Inventor: **Ioan Cosmescu**, Phoenix, AZ (US)
(73) Assignee: **I.C. Medical, Inc.**, Phoenix, AZ (US)
(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 229 days.

4,232,676	A	11/1980	Herczog
4,823,791	A	4/1989	D'Amelio et al.
4,850,353	A	7/1989	Stasz et al.
5,013,312	A	5/1991	Parins et al.
5,171,311	A	12/1992	Rydell et al.
5,277,114	A	1/1994	Witt
5,908,420	A	6/1999	Parins et al.
6,432,104	B1	8/2002	Durgin et al.
8,177,783	B2	5/2012	Davison et al.
2003/0125732	A1	7/2003	Goble
2005/0113825	A1	5/2005	Cosmescu
2006/0025757	A1	2/2006	Heim
2008/0262489	A1	10/2008	Steinke
2009/0306659	A1	12/2009	Buyssse
2014/0276795	A1	9/2014	Batchelor et al.

(21) Appl. No.: **17/929,334**

(22) Filed: **Sep. 2, 2022**

(65) **Prior Publication Data**

US 2022/0409262 A1 Dec. 29, 2022

Related U.S. Application Data

(63) Continuation of application No. 16/850,367, filed on Apr. 16, 2020, now Pat. No. 11,534,237, which is a continuation of application No. 15/211,270, filed on Jul. 15, 2016, now Pat. No. 10,722,299.

(51) **Int. Cl.**
A61B 18/12 (2006.01)
A61B 18/14 (2006.01)
A61B 18/00 (2006.01)

(52) **U.S. Cl.**
CPC **A61B 18/149** (2013.01); **A61B 18/148** (2013.01); **A61B 2018/00607** (2013.01); **A61B 2018/00958** (2013.01); **A61B 2018/1467** (2013.01); **A61B 18/1477** (2013.01)

(58) **Field of Classification Search**
CPC A61B 18/1402; A61B 18/1477; A61B 18/148; A61B 18/149; A61B 2018/00607; A61B 2018/00958; A61B 2018/1467; A61B 2018/162

See application file for complete search history.

OTHER PUBLICATIONS

International Search Report and the Written Opinion of the International Searching Authority from related PCT Application No. PCT/US2017/031137 mailed out Jul. 25, 2017.

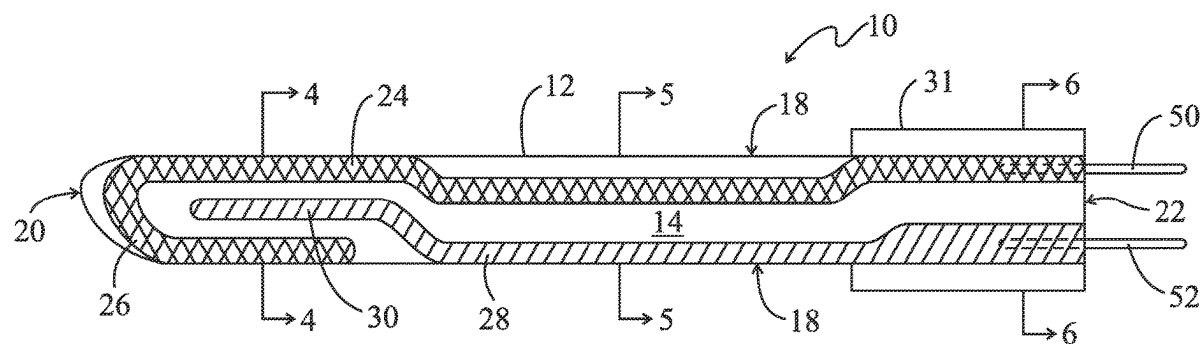
Primary Examiner — Daniel W Fowler

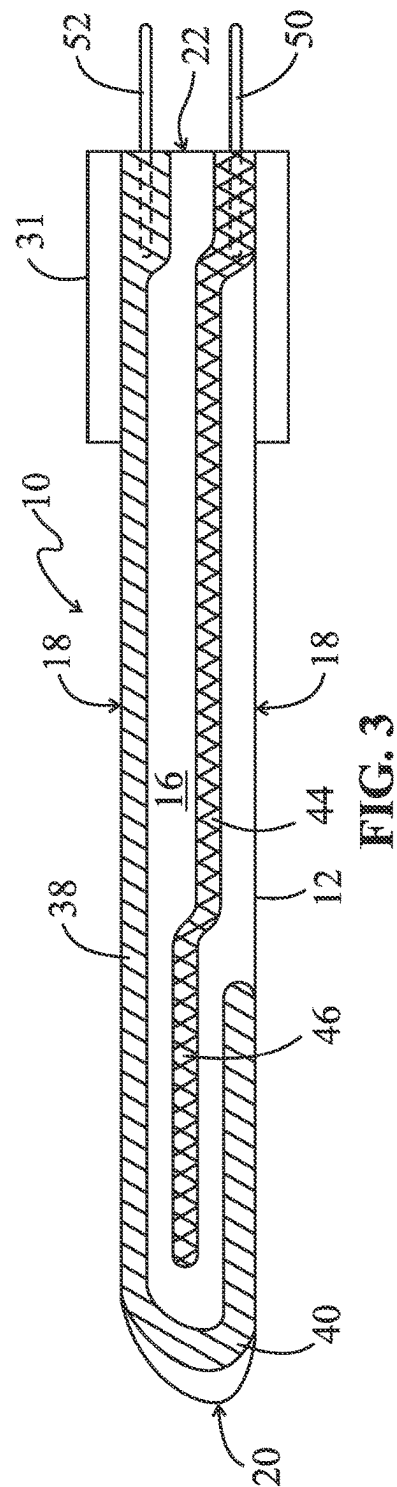
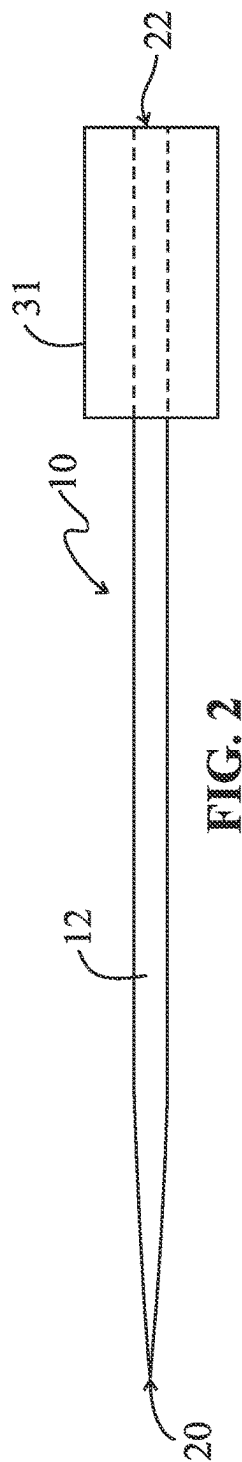
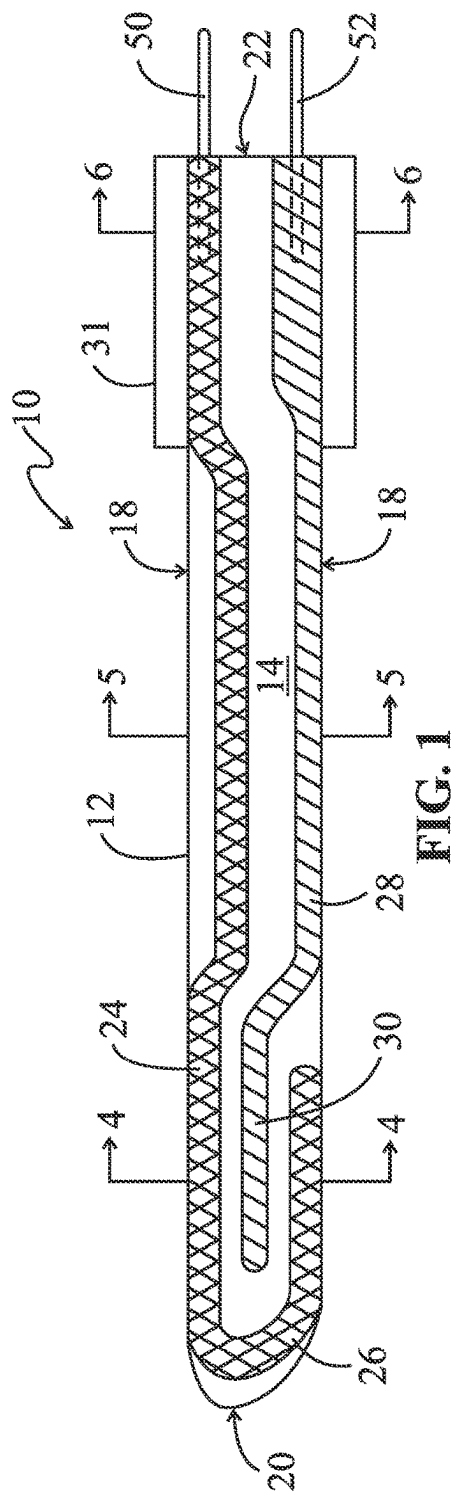
(74) *Attorney, Agent, or Firm* — Zeman-Mullen & Ford, LLP

(57) **ABSTRACT**

An ultrapolar electrosurgery blade and an ultrapolar electrosurgery pencil. The ultrapolar electrosurgery blade has a non-conductive planar member with opposing planar sides, a cutting end, and an opposite non-cutting end, first active and return electrodes located on one opposing planar side, and second active and return electrodes located on the other opposing planar side.

16 Claims, 4 Drawing Sheets





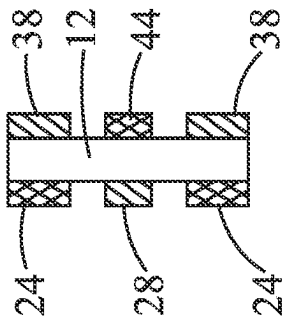


FIG. 4

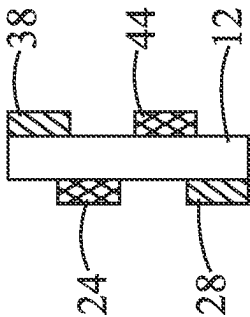


FIG. 5

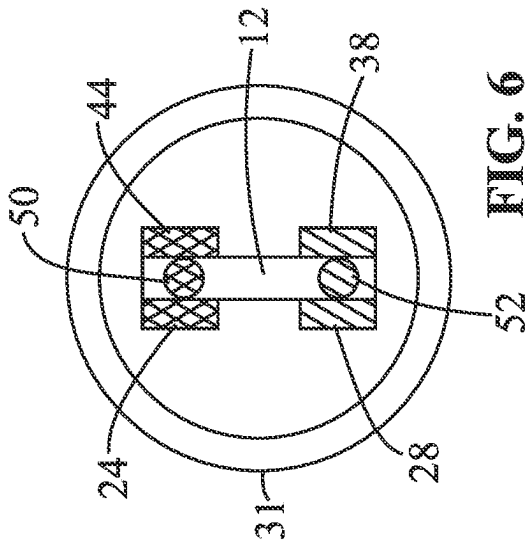


FIG. 6

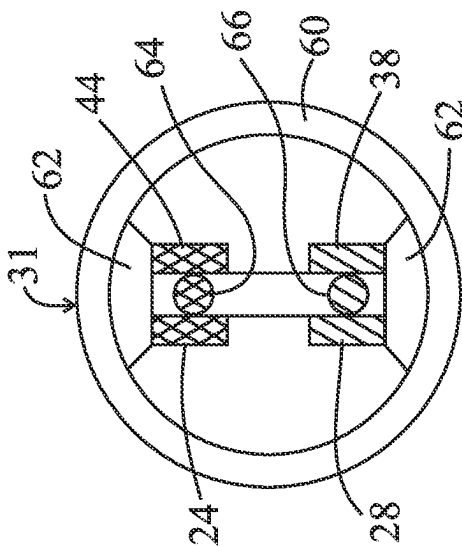


FIG. 7

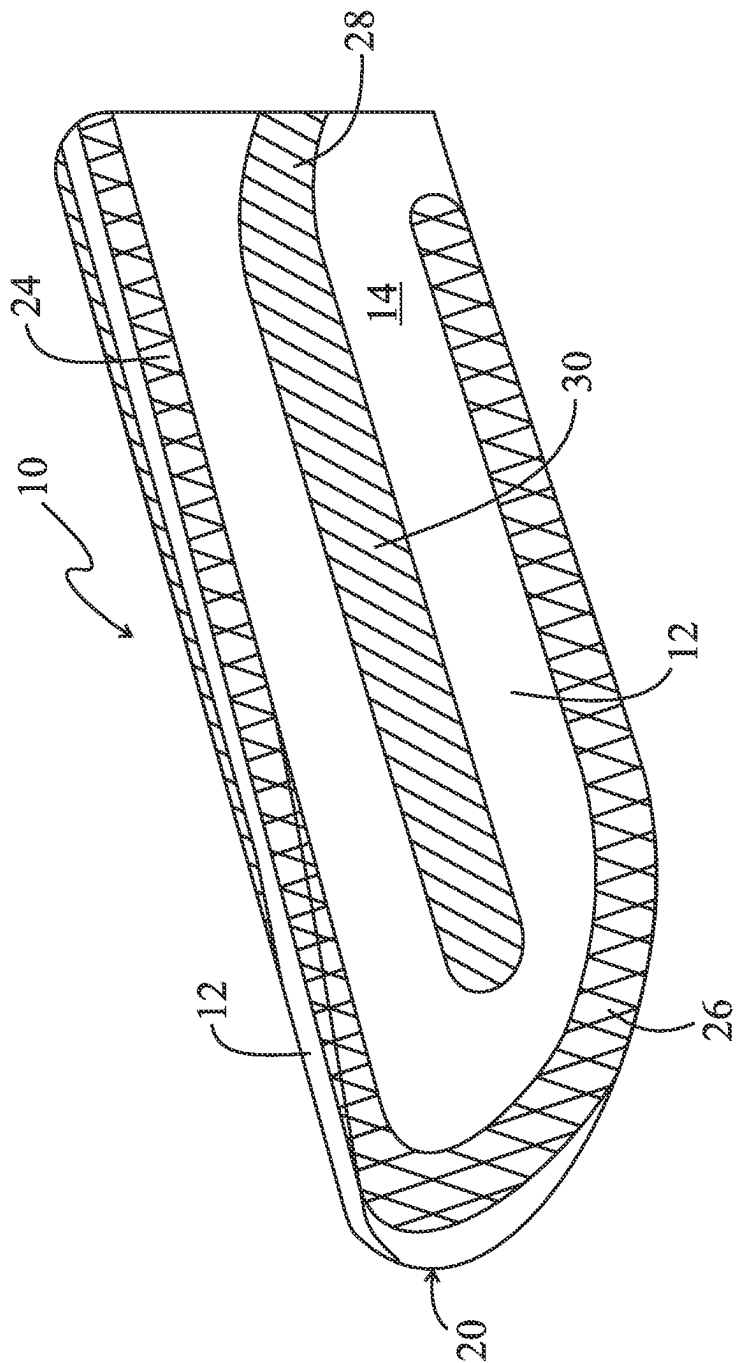


FIG. 8

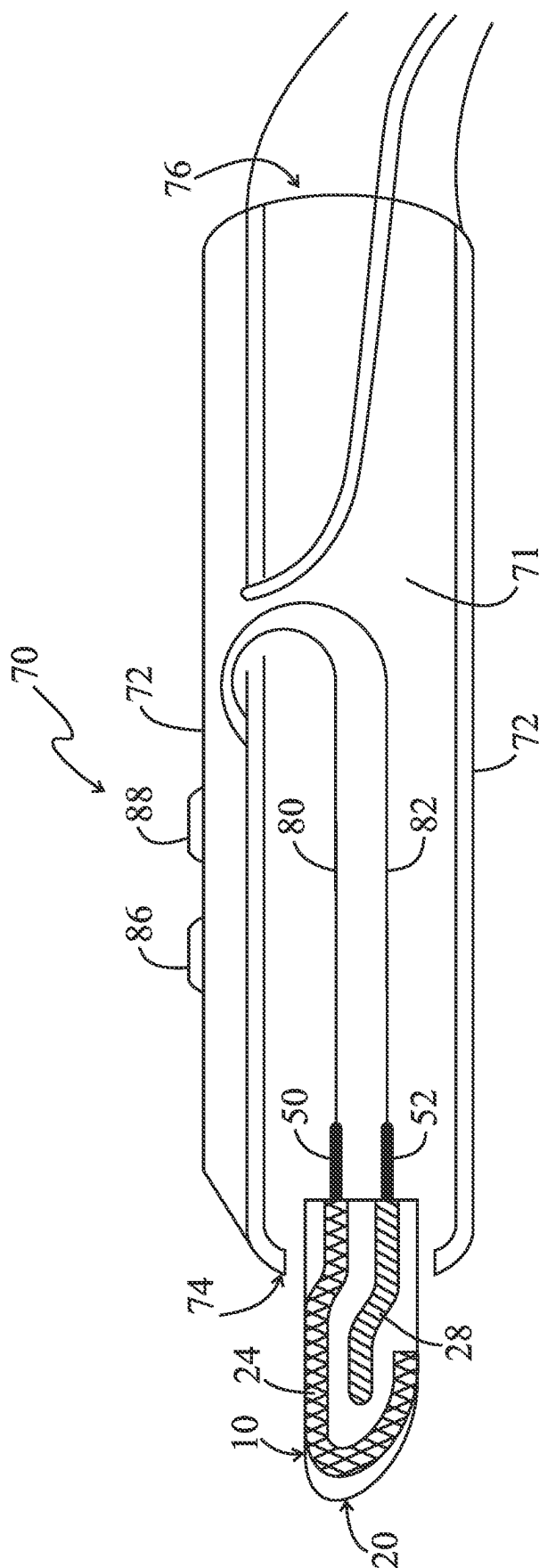


FIG. 9

1

ULTRAPOLAR ELECTROSURGERY BLADE AND ULTRAPOLAR ELECTROSURGERY PENCIL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of, and claims priority to, patent application having Ser. No. 16/850,367 entitled “Ultrapolar Electrosurgery Blade and Ultrapolar Electrosurgery Pencil,” filed Apr. 16, 2020, which is a continuation of, and claims priority to, patent application having Ser. No. 15/211,270, entitled “Ultrapolar Electrosurgery Blade and Ultrapolar Electrosurgery Pencil,” filed Jul. 15, 2016, which are herein incorporated by reference in their entireties.

FIELD OF INVENTION

The present invention is generally directed to an ultrapolar electrosurgery blade and an ultrapolar electrosurgery pencil which use monopolar energy in a bipolar mode for cutting and coagulation. The ultrapolar electrosurgery blade has a non-conductive planar member with first and second opposite planar sides, a cutting end, and an opposite non-cutting end, a first active electrode and a first return electrode both located on the first opposite planar side of the non-conductive planar member, and a second active electrode and a second return electrode both located on the second opposite planar side of the non-conductive planar member.

BACKGROUND OF THE INVENTION

Electrosurgery uses an RF electrosurgical generator (also known as an electrosurgical unit or ESU) and a handpiece with an electrode to provide high frequency, alternating radio frequency (RF) current input at various voltages to cut or coagulate biological tissue. The handpiece may be a monopolar instrument with one electrode or a bipolar instrument with two electrodes. When using a monopolar instrument, a return electrode pad is attached to the patient and the high frequency electrical current flows from the generator, to the monopolar instrument, through the patient to the patient return electrode pad, and back to the generator. Monopolar electrosurgery is commonly used due to its versatility and effectiveness. However, the excessive heat generated with monopolar electrosurgery can cause excessive tissue damage and necrosis of the tissue because the return electrode positioned on the back of the patient causes high voltage and high RF energy to pass through the patient.

In bipolar electrosurgery, active output and patient return functions both occur at the surgery site because both the active and return electrodes are contained in the bipolar instrument. Therefore, the path of the electrical current is confined to the biological tissue located between the active and return electrodes. Although bipolar electrosurgery enables the use of lower voltages and less energy and thereby reduces or eliminates the likelihood of tissue damage and sparking associated with monopolar electrosurgery, it has limited ability to cut and coagulate large bleeding areas.

Accordingly, there is a need for an electrosurgery blade that allows for both cutting and coagulation of large areas of tissue without the tissue damage and which eliminates passing of energy through the patient. An ultrapolar electrosurgery blade having a sharp cutting edge and both active and return electrodes positioned on opposing sides of the

2

electrosurgery blade would meet this need. The ultrapolar electrosurgery blades with a sharp cutting edge and active and return electrodes positioned on both opposing sides of the electrosurgery blade described with reference to the present invention could be used with an electrosurgery handpiece/pencil that does not have smoke evacuation capability but they are also intended to be used with an electrosurgery pencil/handpiece that is capable of smoke evacuation during the electrosurgery procedure.

SUMMARY OF THE INVENTION

The present invention is directed to an ultrapolar electrosurgery blade which includes a non-conductive planar member having first and second opposite planar sides, a cutting end, and a non-cutting end, first active and return electrodes each located on the first opposite planar side wherein at least a portion of the first opposite planar side is exposed near the cutting end of the non-conductive planar member, and second active and return electrodes each located on the second opposite planar side wherein at least a portion of the second opposite planar side is exposed near the cutting end of the non-conductive planar member. In one exemplary embodiment of the ultrapolar electrosurgery blade of the present invention, the first active electrode located on the first planar side mirrors at least a portion of the second return electrode located on the second planar side and the first return electrode located on the first planar side mirrors at least a portion of the second active electrode located on the second planar side.

The non-conductive planar member may comprise a ceramic and the first and second active electrodes and the first and second return electrodes may comprise a stainless steel, a copper, and/or a tungsten. In another exemplary embodiment of the ultrapolar electrosurgery blade of the present invention, the first and second active electrodes and the first and second return electrodes may each take the form of an elongated conductive layer that extends more than half the length of the non-conductive planar member. At least a portion of the elongated conductive layer of the first active electrode may extend to, and along a partial length of, one or more of the opposing elongated edges of the first opposite planar side and at least a portion of the elongated conductive layer of the first return electrode may extend to, and along a partial length of, one or more of the opposing elongated edges of the first opposite planar side.

In still another exemplary embodiment of the ultrapolar electrosurgery blade of the present invention, at least a portion of the elongated conductive layer of the second return electrode on the second opposite planar side of the non-conductive planar member mirrors at least a portion of the elongated conductive layer of the first active electrode on the first opposite planar side of the non-conductive planar member and at least a portion of the elongated conductive layer of the second active electrode on the second opposite planar side of the non-conductive planar member mirrors at least a portion of the elongated conductive layer of the first return electrode on the first opposite planar side of the non-conductive planar member.

In yet another exemplary embodiment of the ultrapolar electrosurgery blade of the present invention, the ultrapolar electrosurgery blade includes a non-conductive planar member having first and second opposite planar sides with opposing elongated edges, a cutting end, and an opposite non-cutting end, a first active electrode having a hook-shaped configuration located on the first opposite planar side of the non-conductive planar member, a first return electrode

3

having a bar-shaped configuration with at least a portion of the first return electrode positioned within the hook-shaped configuration of the first active electrode without touching the first active electrode, a second return electrode having a hook-shaped configuration located on the second opposite planar side of the non-conductive planar member, and a second active electrode having a bar-shaped configuration with at least a portion of the second active electrode positioned within the hook-shaped configuration of the second return electrode without touching the second return electrode. The hook-shaped configuration of the first active electrode and the second return electrode may be located near the cutting end of the non-conductive planar member without covering at least a portion of the first and second opposite planar sides located near the cutting end of the non-conductive planar member. At least a portion of the hook-shaped configuration of the first active electrode located on the first opposite planar side may mirror at least a portion of the hook-shaped configuration of the second return electrode located on the second opposite planar side and at least a portion of the bar-shaped configuration of the first return electrode may mirror at least a portion of the bar-shaped configuration of the second active electrode. The non-conductive planar member may comprise a ceramic and the first and second active electrodes and the first and second return electrodes may comprise a stainless steel, a copper, and/or a tungsten.

The first active electrode may take the form of a hook-shaped conductive layer where at least a portion of the hook-shaped conductive layer extends to, and along a partial length of, the opposing elongated edges of the first opposite planar side of the non-conductive planar member and the first return electrode may take the form of a bar-shaped conductive layer where at least a portion of the bar-shaped conductive layer extends to, and along a partial length of, one of the opposing elongated edges of the first opposite planar side of the non-conductive planar member. Similarly, the second return electrode may take the form of a hook-shaped conductive layer where at least a portion of the hook-shaped conductive layer extends to, and along a partial length of, the opposing elongated edges of the second opposite planar side of the non-conductive planar member and the second active electrode may take the form of a bar-shaped conductive layer where at least a portion of the bar-shaped conductive layer extends to, and along a partial length of, one of the opposing elongated edges of the second opposite planar side of the non-conductive planar member.

Further, a portion of the first and second active electrodes and a portion of the first and second return electrodes may extend to the opposite non-cutting end of the non-conductive planar member. The ultrapolar electrosurgery blade of the present invention may further comprise a first conductive insert member in communication with both the first active electrode and second active electrode located near the non-cutting end of the non-conductive planar member and a second conductive insert member in communication with both the first return electrode and the second return electrode located near the non-cutting end of the non-conductive planar member. The first and second conductive inserts may each comprise a metal contact member that is made of brass and/or copper.

The present invention is also directed to an ultrapolar electrosurgery pencil having a handpiece with a first and second end and an electrosurgery blade positioned in the first end of the handpiece where the electrosurgery blade includes a non-conductive planar member having first and second opposite planar sides, a cutting end, and an opposite

4

non-cutting end, first active and return electrodes located on the first opposite planar side where at least a portion of the first opposite planar side is exposed near the non-conductive cutting end of the non-conductive planar member, and second active and return electrodes located on the second opposite planar side where at least a portion of the second opposite planar side is exposed near the non-conductive cutting end of the non-conductive planar member. The handpiece may further comprise a smoke evacuation channel contained therein for evacuating smoke and debris from the surgical site during activation of the electrosurgery pencil. In addition, the handpiece may further comprise at least one activation button for cutting and at least one activation button for coagulation.

BRIEF DESCRIPTION OF THE DRAWINGS

The subject invention will hereafter be described in conjunction with the appended drawing figures, wherein like numerals denote like elements, and

FIG. 1 is a side view of an exemplary embodiment of the ultrapolar electrosurgery blade of the present invention;

FIG. 2 is a top plan view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 1;

FIG. 3 is an opposite side view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 1 with the ultrapolar electrosurgery blade in FIG. 1 shown rotated 180 degrees;

FIG. 4 is a cross-sectional view taken along line 4-4 of FIG. 1;

FIG. 5 is a cross-sectional view taken along line 5-5 of FIG. 1;

FIG. 6 is a cross-sectional view taken along line 6-6 of FIG. 1;

FIG. 7 is an end view of the ultrapolar electrosurgery blade of FIGS. 1 and 3 showing one exemplary embodiment of a support member for retaining the ultrapolar electrosurgery blade of the present invention so that the conductive inserts in communication with the active and return electrodes of the electrosurgery blade can be easily inserted into an electrosurgery pencil;

FIG. 8 is a partial perspective view of the exemplary embodiment of the ultrapolar electrosurgery blade of the present invention shown in FIG. 1; and

FIG. 9 is a side cross-sectional view of an exemplary embodiment of an ultrapolar electrosurgery pencil of the present invention.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

The exemplary embodiments of the ultrapolar electrosurgery blade and the ultrapolar electrosurgery pencil of the present invention enable a user or surgeon to perform cutting with the sharp non-conductive tip of the electrosurgery blade as well as coagulation of large areas of biological tissue with the electrosurgery blade by placing the electrosurgery blade on either of its sides where both active and return electrodes are located. The electrosurgery blade and electrosurgery pencil of the present invention may also perform cutting with the active and return electrodes of the electrosurgery blade. Exemplary embodiments of the ultrapolar electrosurgery blade and the ultrapolar electrosurgery pencil of the present invention include an electrosurgery blade that has a non-conductive planar member with opposite planar sides, a cutting end, and a non-cutting end, first active and return

5

electrodes each located on one opposite planar side of the non-conductive planar member, and second active and return electrodes each positioned on the other opposite planar side of the non-conductive return electrode. The cutting end of the non-conductive planar member can form a sharp non-conductive cutting end for cutting biological tissue while the active and return electrodes located on each side of the non-conductive planar member can be used to perform coagulation as well as cutting of biological tissue.

FIGS. 1 and 3 show opposing side views of an exemplary embodiment of the ultrapolar electrosurgery blade 10 of the present invention which includes a non-conductive planar member 12 having first and second opposite planar sides 14 (see FIG. 1), 16 (see FIG. 2) with opposing elongated edges 18, a cutting end 20, and an opposite non-cutting end 22, a first active electrode 24 and a first return electrode 28 each located on the first opposite planar side 14 of the non-conductive planar member 12 wherein at least a portion of the first opposite planar side 14 is exposed near the non-conductive cutting end 20 of the non-conductive planar member 12, and a second active electrode 44 and a second return electrode 38 each located on the second opposite planar side 16 of the non-conductive planar member 12 wherein at least a portion of the second opposite planar side 16 is exposed near the non-conductive cutting end 20 of the non-conductive planar member 12. First and second active electrodes 24, 44 and first and second return electrodes 28, 38 may each take the form of an elongated conductive layer that extends more than half the length of the non-conductive planar member 12.

As shown in the exemplary embodiment shown in FIGS. 1 and 3, at least a portion of the first active electrode 24 and at least a portion of the first return electrode 28 may extend to, and along a partial length of, at least one of the opposing elongated edges 18 of the first opposite planar side 14 of the non-conductive planar member 12. Further, at least a portion of the second return electrode 38 may extend to, and along a partial length of, at least one of the opposing elongated edges 18 of the second opposite planar side 16 of the non-conductive planar member 12. As further shown in the exemplary embodiment of the ultrapolar electrosurgery blade shown in FIGS. 1 and 3, at least a portion of the first active electrode 24 located on the first opposite planar side 14 may mirror at least a portion of the second return electrode 38 located on the second opposite planar side 16 and at least a portion of the first return electrode 28 located on the first opposite planar side 14 may mirror at least a portion of the second active electrode 44 located on the second opposite planar side 16. Moreover, the first active electrode 24 may include a hook-shaped configuration 26 and the first return electrode 28 may include a bar-shaped configuration 30 where at least a portion of the bar-shaped configuration 30 of the first return electrode 28 is positioned within the hook-shaped configuration 26 of the first active electrode 24. In addition, the second return electrode 38 may include a hook-shaped configuration 40 and the second active electrode 44 may include a bar-shaped configuration 46 where at least a portion of the bar-shaped configuration 46 of the second active electrode 44 is positioned within the hook-shaped configuration 40 of the second return electrode 38.

The non-conductive planar member 12 may comprise a ceramic which can also form a sharp cutting end 20. First and second active electrodes 24, 44 and first and second return electrodes 28, 38 may comprise at least one of stainless steel, copper, and/or tungsten. The hook-shaped configuration 26 of the first active electrode 24 and the

6

hook-shaped configuration 40 of the second return electrode 38 may be located near cutting end 20 of non-conductive planar member 12 while still enabling at least a portion of the first and second opposite sides 14, 16 of the non-conductive planar member 12 to be exposed near the non-conductive cutting end 20. This enables precise cutting to be performed with the sharp non-conductive cutting end 20 of the ultrapolar electrosurgery blade 10.

Ultrapolar electrosurgery blade 10 may also include a support member 31 which may take the form of a variety of configurations as long as it is capable of supporting the non-conductive planar member 12 and its associated active and return electrodes so that the ultrapolar electrosurgery blade 10 may be easily connected to an instrument such as an electrosurgery pencil. Ultrapolar electrosurgery blade 10 may also include a first conductive insert member 50 that is in communication with the first active electrode 24 and the second active electrode 44 near the opposite non-cutting end 22 of the non-conductive planar member 12 and a second conductive insert member 52 that is in communication with the first return electrode 28 and the second return electrode 38 near the opposite non-cutting end 22 of the non-conductive planar member 12.

FIG. 2 is a top view of the ultrapolar electrosurgery blade 10 of the present invention with the non-conductive planar member 12 shown in phantom within the support member 31. FIG. 4 is a cross-sectional view taken along line 4-4 of FIG. 1 which shows non-conductive planar member 12 and first active electrode 24 and first return electrode 28 located on one side (first opposite planar side 14) of non-conductive planar member 12 and second active electrode 44 and second return electrode 38 located on the other side (second opposite planar side 16) of non-conductive planar member 12. As shown in FIG. 4, there are two portions of first active electrode 24 shown on one side of non-conductive planar member 12 and two portions of second return electrode 38 shown on the other side of non-conductive planar member 12 due to where the cross-section was taken. FIG. 5 shown another cross-section of the ultrapolar electrosurgery blade 10 of the present invention taken along line 5-5 of FIG. 1. FIG. 5 shows one portion of first active electrode 24 and one portion of first return electrode 28 on one side of non-conductive planar member 12 and one portion of second active electrode 44 and one portion of second return electrode 38 on the other side of non-conductive planar member 12. FIG. 6 is a cross-sectional view taken along line 6-6 of FIG. 1 which shows non-conductive planar member 12 along with first and second active electrodes 24, 44 and first and second return electrodes 28, 38 suspended within support member 31. FIG. 6 also shows first conductive insert member 50 of ultrapolar electrosurgery blade 10 in communication with first and second active electrodes 24, 44 and second conductive insert member 52 of ultrapolar electrosurgery blade 10 in communication with first and second return electrodes 28, 38.

An end view of the ultrapolar electrosurgery blade 10 of FIGS. 1 and 3 showing one exemplary embodiment of a support member 31 for retaining the ultrapolar electrosurgery blade 10 of the present invention so that the conductive inserts 50, 52 in communication with the active and return electrodes 24, 44, 28, 38 of the electrosurgery blade 10 can be easily inserted into an instrument such as an electrosurgery pencil. In just one exemplary embodiment, support member 31 may take the form of a hollow cylindrical member 60 having a non-conductive insert member 62 with conductive tubular members 64, 66 into which first and second conductive insert members 50, 52 can be placed. It

7

will be understood by those skilled in the art that the support member for retaining the ultrapolar electrosurgery blade **10** of the present invention may take any number of forms or configurations.

FIG. **8** is a partial perspective view of the exemplary embodiment of the ultrapolar electrosurgery blade **10** of the present invention shown in FIG. **1**. As can be seen in FIG. **8**, the active and return electrodes of the blade are placed on opposite planar sides of the non-conductive planar member **12** with the top of non-conductive planar member **12** remaining non-conductive and free of electrodes.

A side cross-sectional view of an exemplary embodiment of an ultrapolar electrosurgery pencil **70** of the present invention is shown in FIG. **9**. Ultrapolar electrosurgery pencil **70** includes a handpiece **72** having a first end **74** and a second **76**, and an ultrapolar electrosurgery blade **10** positioned within the first end **74** of the handpiece. The ultrapolar electrosurgery blade **10** shown in FIG. **9** is like the blade **10** shown in FIGS. **1-3**. Conductive wires **80**, **82** connect the first and second conductive insert members **50**, **52** to the circuit board which enables at least one activation button **86** for cutting and at least one activation button **88** for coagulation. Handpiece **72** may include a smoke evacuation channel **71** to enable evacuation of smoke and/or debris from the surgical site when performing cutting and/or coagulation.

The detailed description of exemplary embodiments of the invention herein shows various exemplary embodiments of the invention. These exemplary embodiments and modes are described in sufficient detail to enable those skilled in the art to practice the invention and are not intended to limit the scope, applicability, or configuration of the invention in any way. Rather, the following disclosure is intended to teach both the implementation of the exemplary embodiments and modes and any equivalent modes or embodiments that are known or obvious to those reasonably skilled in the art. Additionally, all included examples are non-limiting illustrations of the exemplary embodiments and modes, which similarly avail themselves to any equivalent modes or embodiments that are known or obvious to those reasonably skilled in the art.

Other combinations and/or modifications of structures, arrangements, applications, proportions, elements, materials, or components used in the practice of the instant invention, in addition to those not specifically recited, can be varied or otherwise particularly adapted to specific environments, manufacturing specifications, design parameters, or other operating requirements without departing from the scope of the instant invention and are intended to be included in this disclosure.

Unless specifically noted, it is the Applicant's intent that the words and phrases in the specification and the claims be given the commonly accepted generic meaning or an ordinary and accustomed meaning used by those of ordinary skill in the applicable arts. In the instance where these meanings differ, the words and phrases in the specification and the claims should be given the broadest possible, generic meaning. If any other special meaning is intended for any word or phrase, the specification will clearly state and define the special meaning.

The invention claimed is:

1. An ultrapolar electrosurgery blade comprising:
 - a non-conductive planar member having first and second opposite planar sides;
 - a cutting end, and an opposite non-cutting end;
 - a first active electrode and a first return electrode each located on the first opposite planar side of the non-

8

conductive planar member wherein the first active electrode is positioned near the cutting end such that a portion of the first opposite planar side is exposed between the first active electrode and the cutting end and at least a portion of the first active electrode extends along a length of an edge of the first opposite planar side;

- a second active electrode and a second return electrode each located on the second opposite planar side of the non-conductive planar member; and
- a conductive insert member contained within the non-cutting end of the non-conductive planar member wherein the conductive insert member is in physical communication with the first active electrode and the second active electrode near the opposite non-cutting end of the non-conductive planar member.

2. The ultrapolar electrosurgery blade of claim **1** wherein at least a portion of the second opposite planar side is exposed near between the second active electrode and the cutting end of the non-conductive planar member.

3. The ultrapolar electrosurgery pencil of claim **1** further comprising a second conductive insert member contained within the non-cutting end of the non-conductive planar member wherein the conductive insert member is in physical communication with the first return electrode and the second return electrode near the opposite non-cutting end of the non-conductive planar member.

4. The ultrapolar electrosurgery blade of claim **2** wherein at least a portion of the second active electrode extends along a length of an edge of the second opposite planar side.

5. The ultrapolar electrosurgery blade of claim **1** wherein at least a portion of the first active electrode located on the first opposite planar side mirrors at least a portion of the second return electrode located on the second opposite planar side and at least a portion of the first return electrode on the first opposite planar side mirrors at least a portion of the second active electrode on the second opposite planar side.

6. The ultrapolar electrosurgery blade of claim **1** wherein the non-conductive planar member comprises a ceramic.

7. The ultrapolar electrosurgery blade of claim **1** wherein the first and second active electrodes and the first and second return electrodes each comprise at least one of a stainless steel, a copper, and a tungsten.

8. The ultrapolar electrosurgery blade of claim **1** wherein the first active electrode, the first return electrode, the second active electrode, and the second return electrode each comprise an elongated conductive layer that extends more than half the length of the non-conductive planar member.

9. The ultrapolar electrosurgery blade of claim **8** wherein at least a portion of the elongated conductive layer of the first return electrode extends to, and along a partial length of, an edge of the first opposite planar side of the non-conductive planar member.

10. The ultrapolar electrosurgery blade of claim **8** wherein at least a portion of the elongated conductive layer of the second return electrode on the second opposite planar side of the non-conductive planar member mirrors at least a portion of the elongated conductive layer of the first active electrode on the first opposite planar side of the non-conductive planar member and at least a portion of the elongated conductive layer of the second active electrode on the second opposite planar side of the non-conductive planar member mirrors at least a portion of the elongated conductive layer of the first return electrode on the first opposite planar side of the non-conductive planar member.

9

11. An ultrapolar electrosurgery blade comprising:
 a non-conductive planar member having first and second
 opposite planar sides with opposing elongated edges, a
 cutting end, and an opposite non-cutting end;
 a first hook-shaped active electrode located on the first
 opposite planar side of the non-conductive planar mem-
 ber near the cutting end such that a portion of the first
 opposite planar side is exposed between the cutting end
 and the first hook-shaped active electrode and at least
 a portion of the first hook-shaped active electrode
 extends along a length of one of the opposing elongated
 edges of the first opposite planar side;
 a first return electrode comprising a bar-shaped configu-
 ration wherein a bar-shaped portion of the first return
 electrode is positioned within the hook-shaped configu-
 ration of the first active electrode without touching the
 first active electrode;
 a second hook-shaped return electrode located on the
 second opposite planar side of the non-conductive
 planar member; and
 a second active electrode comprising a bar-shaped con-
 figuration wherein a bar shaped portion of the second
 active electrode is positioned within the hook-shaped
 configuration of the second return electrode without
 touching the second return electrode.

12. The ultrapolar electrosurgery blade of claim **11**
 wherein the second hook-shaped return electrode is located
 near the cutting end of the non-conductive planar member
 such that a portion of second opposite planar side is exposed
 between the cutting end and the second hook-shaped return
 electrode.

10

13. The ultrapolar electrosurgery blade of claim **12**
 wherein the second active electrode comprises a bar-shaped
 conductive layer where at least a portion of the bar-shaped
 conductive layer extends to, and along a length of, one of the
 opposing elongated edges of the second opposite planar side
 of the non-conductive planar member.

14. The ultrapolar electrosurgery blade of claim **11**
 wherein at least a portion of the first hook-shaped active
 electrode located on the first opposite planar side mirrors at
 least a portion of the second hook-shaped return electrode
 located on the second opposite planar side and at least a
 portion of the bar-shaped configuration of the first return
 electrode located on the first opposite planar side mirrors at
 least a portion of the bar-shaped configuration of the second
 active electrode located on the second opposite planar side.

15. The ultrapolar electrosurgery blade of claim **11**
 wherein a portion of the first and second active electrodes
 and a portion of the first and second return electrodes extend
 to the opposite non-cutting end of the non-conductive planar
 member.

16. The ultrapolar electrosurgery blade of claim **11** further
 comprising a first conductive insert member in communi-
 cation with the first active electrode and second active
 electrode located near the opposite non-cutting end of the
 non-conductive planar member and a second conductive
 insert member in communication with the first return elec-
 trode and the second return electrode located near the
 opposite non-cutting end of the non-conductive planar mem-
 ber.

* * * * *